

1. VEM spaces. Let $\Omega \subset \mathbb{R}^d$, $d = 2, 3$ be a bounded and simply connected polytopal domain. Given an external force field $\mathbf{f} \in \mathbf{L}^2(\Omega)$, we consider the steady-state incompressible Stokes problem:

$$(1.1) \quad \begin{cases} -\operatorname{div}(\varepsilon(\mathbf{u})) + \nabla p = \mathbf{f} & \text{in } \Omega, \\ \operatorname{div} \mathbf{u} = 0 & \text{in } \Omega, \\ \mathbf{u} = \mathbf{0} & \text{on } \partial\Omega. \end{cases}$$

1.1. Primal and dual meshes. Let $\mathcal{K}_h$ be a shape-regular polytopal mesh of the domain Ω . We assume each element $K \in \mathcal{K}_h$ is a star-shaped polytope with a bounded chunkiness parameter. Let $\mathcal{F}_h$ be the set of $(d-1)$ -dimensional faces of $\mathcal{K}_h$, and $\mathring{\mathcal{F}}_h^K = \mathcal{F}_h \setminus \partial\Omega$. We call $\mathcal{K}_h$ the primal mesh.

For each $K \in \mathcal{K}_h$, let $\mathbf{x}_K$ be the center of the largest ball inside K . Connecting $\mathbf{x}_K$ to the vertices of K gives a shape-regular refinement $\mathcal{T}_h$. Let $\mathcal{T}_h(K)$ be the set of pyramids in K . Let $\mathcal{F}_h^T$ be the set of $(d-1)$ -dimensional faces of $\mathcal{T}_h$, and $\mathring{\mathcal{F}}_h^T = \mathcal{F}_h^T \setminus \partial\Omega$.

Let $F \in \partial K$ be a face with unit normal $\mathbf{n}_F$ and orthonormal tangent vectors $\{\mathbf{t}_i^F\}_{i=1}^{d-1}$. For any $\mathbf{v} \in \mathbb{R}^d$, the tangential part on F is

$$\Pi_F \mathbf{v} = (\mathbf{I} - \mathbf{n}_F \otimes \mathbf{n}_F) \mathbf{v} = \sum_{i=1}^{d-1} (\mathbf{v} \cdot \mathbf{t}_i^F) \mathbf{t}_i^F.$$

Define the following spaces on F :

$$\begin{aligned} \mathcal{T}^F &:= \operatorname{span}\{\mathbf{t}_1^F, \dots, \mathbf{t}_{d-1}^F\}, \\ \mathcal{T}^F(\mathbb{S}) &:= \operatorname{span}\{\operatorname{sym}(\mathbf{t}_i^F \otimes \mathbf{t}_j^F) \mid 1 \leq i \leq j \leq d-1\}, \\ \mathcal{N}^F(\mathbb{S}) &:= \operatorname{sym}(\mathcal{T}^F \otimes \mathbf{n}_F) \oplus \operatorname{span}\{\mathbf{n}_F \otimes \mathbf{n}_F\}, \\ \mathcal{T}^F(\mathbb{K}) &:= \operatorname{span}\{\operatorname{skw}(\mathbf{t}_i^F \otimes \mathbf{t}_j^F) \mid 1 \leq i < j \leq d-1\}. \end{aligned}$$

These subspaces give the following $\mathbf{t}\text{-}\mathbf{n}$ decompositions

$$(1.2) \quad \mathbb{S} = \mathcal{T}^F(\mathbb{S}) \oplus \mathcal{N}^F(\mathbb{S}),$$

$$(1.3) \quad \mathbb{K} = \mathcal{T}^F(\mathbb{K}) \oplus \mathcal{N}^F(\mathbb{K}), \quad \mathcal{N}^F(\mathbb{K}) = \operatorname{skw}(\mathcal{T}^F \otimes \mathbf{n}_F).$$

For $K \in \mathcal{K}_h$, let $\mathbb{P}_k(K)$ be the space of polynomials on K of degree at most k . Let $\mathbb{P}_k(K; \mathbb{X}) = \mathbb{P}_k(K) \otimes \mathbb{X}$, where $\mathbb{X}$ is $\mathbb{R}^d$ for vectors, $\mathbb{S}$ for symmetric matrices, and $\mathbb{K}$ for skew-symmetric matrices.

1.2. VEM spaces. On each element $K \in \mathcal{K}_h$, we introduce a local space

$$\mathbf{V}_K = \left\{ \mathbf{v} \in \mathbf{H}(\operatorname{div}, K) \cap \mathbf{H}(\operatorname{curl}, K) : \mathbf{v} \cdot \mathbf{n} \in \mathbb{P}_k(F) \ \forall F \subset \partial K, \right. \\ \left. \operatorname{div} \mathbf{v} \in \mathbb{P}_{k-1}(K), \operatorname{curl} \mathbf{v} \in \operatorname{curl} \mathbb{P}_k(K; \mathbb{R}^d) \right\},$$

The global virtual element space is

$$\mathbf{V}_h^{\operatorname{div}} := \{\mathbf{v} \in \mathbf{H}(\operatorname{div}, \Omega), \mathbf{v}|_K \in \mathbf{V}_K \ \forall K \in \mathcal{K}_h, \mathbf{v} \cdot \mathbf{n} = 0 \text{ on } \partial\Omega\}.$$

The degree of freedoms (d.o.f.) for functions in $\mathbf{V}_K$ are stated as

$$\text{Type I} \quad \int_F \mathbf{v} \cdot \mathbf{n} q_k \, ds \quad \forall q_k \in \mathbb{P}_k(F), F \subset \partial K,$$

$$\text{Type II} \quad \int_K \mathbf{v} \cdot \mathbf{q}_{k-2} \, d\mathbf{x} \quad \forall \mathbf{q}_{k-2} \in \mathbf{G}_{k-2}(K),$$

$$\text{Type III} \quad \int_K \mathbf{v} \cdot \mathbf{q}_k \, d\mathbf{x} \quad \forall \mathbf{q}_k \in \mathbb{P}_k(K; \mathbb{R}^d) \setminus \mathbf{G}_k(K),$$

where $\mathbf{G}_k(K)$ stands for the range of the gradient of the polynomial space $\mathbb{P}_{k+1}(K)$.

On each face $F \in \mathcal{F}_h$, we introduce a space

$$\mathbf{V}_F := \mathbb{P}_{k-1}(F; \mathcal{T}^F) + \mathbf{RM}(F),$$

where $\mathbf{RM}(F)$ denotes the space of rigid motions, i.e., $\mathbf{RM}(F) := \Pi_F \mathbf{RM}(K) = \mathbb{P}_0(F; \mathcal{T}^F) \oplus \mathbb{P}_0(F; \mathcal{T}^F(\mathbb{K}))\mathbf{x}$. The d.o.f. for functions in $\mathbf{V}_F$ are

$$\int_F \mathbf{v} \cdot \mathbf{q} \, ds \quad \forall \mathbf{q} \in \mathbb{P}_{k-1}(F; \mathcal{T}^F) + \mathbf{RM}(F).$$

Then, the element boundary space is defined as

$$\mathbf{V}_h^t := \{\mathbf{v}^t, \mathbf{v}^t|_F \in \mathbf{V}_F, F \in \mathcal{F}_h, \mathbf{v}^t = \mathbf{0} \text{ on } \partial\Omega\}.$$

Combining the above two kinds of spaces together, gives our weak virtual finite element space:

$$\mathbf{V}_h := \left\{ \{\mathbf{v}_h^{\text{div}}, \mathbf{v}_h^t\}, \mathbf{v}_h^{\text{div}} \in \mathbf{V}_h^{\text{div}}, \mathbf{v}_h^t \in \mathbf{V}_h^t \right\}.$$

1.3. Projection operator. On each $K \in \mathcal{K}_h$, we shall define a L^2 projection. For each $\mathbf{v} \in \mathbf{V}_K$, let $\Pi_K^o \mathbf{v} \in \mathbb{P}_k(K; \mathbb{R}^d)$ satisfy

$$\int_K \Pi_K^o \mathbf{v} \cdot \mathbf{q} \, d\mathbf{x} := \int_K \mathbf{v} \cdot \mathbf{q} \, d\mathbf{x} \quad \forall \mathbf{q} \in \mathbb{P}_k(K; \mathbb{R}^d).$$

Although the function in $\mathbf{V}_K$ is virtual, the L^2 projection is computable. In view of Type III d.o.f., we only need to determine $\int_K \mathbf{v} \cdot \nabla p_{k+1}$ for $p_{k+1} \in \mathbb{P}_{k+1}(K)$ which is again computable through integration by parts

$$\int_K \mathbf{v} \cdot \nabla p_{k+1} \, d\mathbf{x} = - \int_K \text{div } \mathbf{v} p_{k+1} \, d\mathbf{x} + \int_{\partial K} \mathbf{v} \cdot \mathbf{n} p_{k+1} \, ds.$$

Note that although $p_{k+1} \in \mathbb{P}_{k+1}(K)$, the polynomials $\text{div } \mathbf{v}$ and $\mathbf{v} \cdot \mathbf{n}$ have been computed and considered as known.

2. Weak strain.

2.1. Weak strain space. For $T \in \mathcal{T}_h$, the local weak strain space is defined as

$$\begin{aligned} \Sigma_k(T) &= \text{sym}(\mathbf{n}_F \otimes V_k(F)) + \mathbb{P}_{k-1}(T; \mathbb{S}) \\ &= (\mathbf{n}_F \otimes \mathbf{n}_F \mathbb{P}_k(T)) \oplus \text{sym}(\mathbf{n}_F \otimes (\mathbb{P}_{k-1}(T; \mathcal{T}^F) + \mathbf{RM}(F))) \oplus \mathbb{P}_{k-1}(T; \mathcal{T}^F(\mathbb{S})), \end{aligned}$$

where $F \in \mathcal{F}_h^K \cap \partial T$ and

$$V_k(F) := \mathbb{P}_k(F) \mathbf{n}_F \oplus \mathbf{V}_F.$$

53 For $k = 1$, we have

$$54 \quad \Sigma_1(T) = (\mathbf{n}_F \otimes \mathbf{n}_F \mathbb{P}_1(T)) \oplus \text{sym}(\mathbf{n}_F \otimes \mathbf{RM}(F)) \oplus \mathbb{P}_0(T; \mathcal{T}^F(\mathbb{S})).$$

55 For $k > 1$, we get

$$56 \quad \Sigma_k(T) = (\mathbf{n}_F \otimes \mathbf{n}_F \mathbb{P}_k(T)) \oplus \text{sym}(\mathbf{n}_F \otimes \mathbb{P}_{k-1}(T; \mathcal{T}^F)) \oplus \mathbb{P}_{k-1}(T; \mathcal{T}^F(\mathbb{S})).$$

57 Note that, we naturally extend the domain of polynomials in $V_k(F)$ from F to
 58 K (or to a sub-element $T \in \mathcal{T}_h(K)$ sharing F) by assuming the function is constant
 59 along lines perpendicular to F .

60 The trace of $\Sigma_k(T)$ onto the face $F \in \mathcal{F}_h^K \cap \partial T$ is then given by

$$61 \quad \text{tr}_F(\Sigma_k(T)) = V_k(F).$$

62 We define the local symmetric tensor space on K as the discontinuous polynomial
 63 space

$$64 \quad (2.1) \quad \Sigma_h^{-1}(K) = \bigoplus_{i=1}^n \chi_{T_i} \Sigma_k(T_i) = \{\boldsymbol{\tau} \in L^2(K; \mathbb{S}) : \boldsymbol{\tau}|_T \in \Sigma_k(T) \text{ for } T \in \mathcal{T}_h(K)\},$$

65 where χ_{T_i} is the characteristic function of T_i . We define the bubble space

$$\begin{aligned} \mathbb{B}_k(\text{div}, K; \mathbb{S}) = & \{\boldsymbol{\tau} \in \Sigma_h^{-1}(K) : \text{tr } \boldsymbol{\tau} := \boldsymbol{\tau} \mathbf{n}|_{\partial K} = \mathbf{0}\} \\ & = \bigoplus_{i=1}^n \lambda_{\mathbf{x}_K} \chi_{T_i} (\mathbb{P}_{k-1}(T_i) \mathbf{n}_i \otimes \mathbf{n}_i) \\ & \oplus \bigoplus_{i=1}^n \lambda_{\mathbf{x}_K} \chi_{T_i} \text{sym}(\mathbb{P}_{k-2}(T_i; \mathcal{T}^F) \otimes \mathbf{n}_i) \\ & \oplus \bigoplus_{i=1}^n \chi_{T_i} \mathbb{P}_{k-1}(T_i; \mathcal{T}^{F_i}(\mathbb{S})). \end{aligned}$$

66 (2.2)

67 *Assumption 2.1.* There exist $d(d+1)/2$ symmetric tensors b_ℓ such that

$$68 \quad (2.3) \quad \mathbb{S} = \text{span}\{b_\ell, \ell = 1, \dots, d(d+1)/2\},$$

69 where $\mathcal{T}^{F_i}(\mathbb{S}) = \text{span}\{b_i^m, m = 1, \dots, d(d-1)/2\}$ for $F_i \in \partial K$, and

$$70 \quad (2.4) \quad \{b_\ell\} \subset \{b_i^m, i = 1, \dots, n, m = 1, \dots, d(d-1)/2\}.$$

71 We now formally define the DoFs for the local stress space $\Sigma_h^{-1}(K)$.

72 **LEMMA 2.2.** *For $K \in \mathcal{K}_h$, the space $\Sigma_h^{-1}(K)$ is uniquely determined by the fol-*
 73 *lowing DoFs:*

$$74 \quad (2.5a) \quad \int_F \boldsymbol{\sigma} \mathbf{n} \cdot \mathbf{q} \, ds, \quad \mathbf{q} \in V_k(F), F \in \partial K,$$

$$75 \quad (2.5b) \quad \int_K \boldsymbol{\sigma} : \boldsymbol{\tau} \, dx, \quad \boldsymbol{\tau} \in \mathbb{B}_k(\text{div}, K; \mathbb{S}) \setminus \bigoplus_{\ell=1}^{d(d+1)/2} \left(\chi_{T_{\mathbf{e}_\ell}} \mathbb{P}_{k-1}(T_{\mathbf{e}_\ell}) b_\ell \right),$$

$$76 \quad (2.5c) \quad \int_K \boldsymbol{\sigma} : \boldsymbol{\tau} \, dx, \quad \boldsymbol{\tau} \in \mathbb{P}_{k-1}(K; \mathbb{S}),$$

77 where $T_{\mathbf{e}_\ell} \in \mathcal{T}_h(K)$ is a sub-element containing the face basis b_ℓ chosen in (2.4).

78 Let

$$79 \quad \Sigma_h^{-1} = \{\boldsymbol{\sigma} \in L^2(\Omega; \mathbb{S}) : \boldsymbol{\sigma}|_T \in \Sigma_k(T), T \in \mathcal{T}_h\}.$$

80 **2.2. Weak strain.** For any $\mathbf{v} = \{\mathbf{v}^{\text{div}}, \mathbf{v}^t\} \in \mathbf{V}_h$, we define $\epsilon_h^w(\mathbf{v}) \in \Sigma_h^{-1}$ sat-
 81 isfying $\epsilon_h^w(\mathbf{v})|_K = \epsilon_K^w(\mathbf{v})$ and

$$\begin{aligned} & (\epsilon_K^w(\mathbf{v}), \boldsymbol{\sigma})_K \\ & := (\epsilon(\boldsymbol{\Pi}_K^o \mathbf{v}^{\text{div}}), \boldsymbol{\sigma})_K + \langle (\mathbf{v}^{\text{div}} - \boldsymbol{\Pi}_K^o \mathbf{v}^{\text{div}}) \cdot \mathbf{n}, (\boldsymbol{\sigma} \mathbf{n}) \cdot \mathbf{n} \rangle_{\partial K} \\ & + \sum_{i=1}^{d-1} \langle (\mathbf{v}^t - \boldsymbol{\Pi}_K^o \mathbf{v}^{\text{div}}) \cdot \boldsymbol{\tau}_i, (\boldsymbol{\sigma} \mathbf{n}) \cdot \boldsymbol{\tau}_i \rangle_{\partial K}, \end{aligned} \quad (2.6)$$

83 for all $\boldsymbol{\sigma} \in \Sigma_h^{-1}(K)$.

84 Note that since $\boldsymbol{\sigma}$ is discontinuous on K ,

$$\begin{aligned} & (\epsilon_K^w(\mathbf{v}), \boldsymbol{\sigma})_K \\ & \neq -(\mathbf{v}^{\text{div}}, \text{div } \boldsymbol{\sigma})_K + \langle \mathbf{v}^{\text{div}} \cdot \mathbf{n}, (\boldsymbol{\sigma} \mathbf{n}) \cdot \mathbf{n} \rangle_{\partial K} + \sum_{i=1}^{d-1} \langle \mathbf{v}^t \cdot \boldsymbol{\tau}_i, (\boldsymbol{\sigma} \mathbf{n}) \cdot \boldsymbol{\tau}_i \rangle_{\partial K} \\ & \neq -(\boldsymbol{\Pi}_K^o \mathbf{v}^{\text{div}}, \text{div } \boldsymbol{\sigma})_K + \langle \mathbf{v}^{\text{div}} \cdot \mathbf{n}, (\boldsymbol{\sigma} \mathbf{n}) \cdot \mathbf{n} \rangle_{\partial K} + \sum_{i=1}^{d-1} \langle \mathbf{v}^t \cdot \boldsymbol{\tau}_i, (\boldsymbol{\sigma} \mathbf{n}) \cdot \boldsymbol{\tau}_i \rangle_{\partial K} \end{aligned} \quad (2.7)$$

86
 87 Let

$$88 \quad \mathcal{J}_{\partial K}^{\mathbf{n}}(\mathbf{v}) := ((\mathbf{v}^{\text{div}} - \boldsymbol{\Pi}_K^o \mathbf{v}^{\text{div}}) \cdot \mathbf{n}) \mathbf{n}, \quad \mathcal{J}_{\partial K}^{\boldsymbol{\tau}_i}(\mathbf{v}) := ((\mathbf{v}^t - \boldsymbol{\Pi}_K^o \mathbf{v}^{\text{div}}) \cdot \boldsymbol{\tau}_i) \boldsymbol{\tau}_i.$$

89 When the superscript omitted, it indicates the whole jump on the element bound-
 90 ary

$$91 \quad \mathcal{J}_{\partial K}(\mathbf{v}) := \mathcal{J}_{\partial K}^{\mathbf{n}}(\mathbf{v}) + \mathcal{J}_{\partial K}^{\boldsymbol{\tau}}(\mathbf{v}) \quad \text{with} \quad \mathcal{J}_{\partial K}^{\boldsymbol{\tau}}(\mathbf{v}) := \sum_{i=1}^{d-1} \mathcal{J}_{\partial K}^{\boldsymbol{\tau}_i}(\mathbf{v}).$$

92 On the face $F \subset \partial K$, we define $(\boldsymbol{\Pi}_{\partial K}^t \mathbf{v})|_F \in \mathbf{V}_F$ as

$$93 \quad \int_F \boldsymbol{\Pi}_{\partial K}^t \mathbf{v} \cdot \mathbf{q} := \int_F \mathbf{v}|_K \cdot \mathbf{q} \quad \forall \mathbf{q} \in \mathbf{V}_F.$$

94 Define the notation $\boldsymbol{\Pi}_{\partial K}$ as

$$95 \quad \boldsymbol{\Pi}_{\partial K} \mathbf{v} := (\mathbf{v} \cdot \mathbf{n}) \mathbf{n} + \boldsymbol{\Pi}_{\partial K}^t \mathbf{v}.$$

96 With these notations, we can simply write (2.6) as

$$97 \quad (\epsilon_K^w(\mathbf{v}), \boldsymbol{\sigma})_K = (\epsilon(\boldsymbol{\Pi}_K^o \mathbf{v}^{\text{div}}), \boldsymbol{\sigma})_K + \langle \boldsymbol{\Pi}_{\partial K} \mathcal{J}_{\partial K}(\mathbf{v}), \boldsymbol{\sigma} \mathbf{n} \rangle_{\partial K}.$$

98 For any $\mathbf{v}_h \in \mathbf{V}_h$, we introduce a mesh dependent $\mathbf{H}^1$ seminorm as

$$99 \quad \|\mathbf{v}_h\|_{1,h}^2 := \sum_{K \in \mathcal{K}_h} \|\mathbf{v}_h\|_{1,h,K}^2,$$

100 where the seminorm on each element is defined as

$$101 \quad \|\mathbf{v}_h\|_{1,h,K}^2 := \|\epsilon(\boldsymbol{\Pi}_K^o \mathbf{v}_h^{\text{div}})\|_K^2 + \|h^{-\frac{1}{2}} \boldsymbol{\Pi}_{\partial K} \mathcal{J}_{\partial K}(\mathbf{v}_h)\|_{\partial K}^2.$$

102 **LEMMA 2.3.** *The mesh-dependent quantity $\|\mathbf{v}_h\|_{1,h}$ defines a norm on the virtual*
 103 *element space $\mathbf{V}_h$.*

Proof. It is clear that $\|\cdot\|_{1,h}$ is a seminorm. We only need to prove positive definiteness. Suppose $\|\mathbf{v}_h\|_{1,h} = 0$ for some $\mathbf{v}_h \in \mathbf{V}_h$. Then for every $K \in \mathcal{K}_h$,

$$\epsilon(\mathbf{\Pi}_K^o \mathbf{v}_h^{\text{div}}) = 0 \quad \text{on } K.$$

and

$$(\mathbf{v}^{\text{div}} - \mathbf{\Pi}_K^o \mathbf{v}^{\text{div}}) \cdot \mathbf{n} = 0, \quad \text{and} \quad \mathbf{v}^t - \mathbf{\Pi}_{\partial K}^t \mathbf{\Pi}_K^o \mathbf{v}^{\text{div}} = \mathbf{0} \quad \text{on } \partial K.$$

The first identity implies that $\mathbf{\Pi}_K^o \mathbf{v}_h^{\text{div}} = \mathbf{r}_K \in \mathbf{RM}(K)$ is a rigid motion vector on each element K . The third identity gives, for every face $F \subset \partial K$ and every tangential vector $\boldsymbol{\tau} \in \mathcal{T}^F$,

$$\mathbf{v}_h^t \cdot \boldsymbol{\tau} = \mathbf{r}_K \cdot \boldsymbol{\tau} \quad \text{on } F.$$

Let F be an interior face shared by two elements K and K' . From the above relation applied on both sides, we obtain

$$(\mathbf{r}_K - \mathbf{r}_{K'}) \cdot \boldsymbol{\tau} = 0, \quad \forall \boldsymbol{\tau} \in \mathcal{T}^F.$$

Since $\mathbf{v}_h^{\text{div}} \in \mathbf{H}(\text{div}, \Omega)$, its normal component is continuous across F . This implies

$$(\mathbf{r}_K - \mathbf{r}_{K'}) \cdot \mathbf{n}_F = 0.$$

Thus $\mathbf{r}_K = \mathbf{r}_{K'}$ on F . By the connectedness of the mesh $\mathcal{K}_h$, there exists a global rigid motion vector $\mathbf{r}$ such that $\mathbf{r}_K = \mathbf{r}$ for all $K \in \mathcal{K}_h$.

Consequently, $\mathbf{\Pi}_K^o \mathbf{v}_h^{\text{div}} = \mathbf{r}$ on every K . Since $\mathbf{r} \cdot \mathbf{n}|_F \in \mathbb{P}_k(F)$, $\text{div } \mathbf{r} = 0 \in \mathbb{P}_{k-1}(K)$, and $\text{curl } \mathbf{r} \in \text{curl } \mathbb{P}_k(K; \mathbb{R}^d)$, we have $\mathbf{r} \in \mathbf{V}_K$. The second identity gives

$$\int_F \mathbf{v}^{\text{div}} \cdot \mathbf{n} q_k \, ds = \int_F \mathbf{r} \cdot \mathbf{n} q_k \, ds \quad \forall q_k \in \mathbb{P}_k(F), F \subset \partial K.$$

The definition of $\mathbf{\Pi}_K^o$

$$\int_K \mathbf{v}^{\text{div}} \cdot \mathbf{q} := \int_K \mathbf{r} \cdot \mathbf{q} \quad \forall \mathbf{q} \in \mathbb{P}_k(K; \mathbb{R}^d)$$

yields

$$\begin{aligned} \int_K \mathbf{v}^{\text{div}} \cdot \mathbf{q}_{k-2} \, d\mathbf{x} &= \int_K \mathbf{r} \cdot \mathbf{q}_{k-2} \, d\mathbf{x} \quad \forall \mathbf{q}_{k-2} \in \mathbf{G}_{k-2}(K), \\ \int_K \mathbf{v}^{\text{div}} \cdot \mathbf{q}_k \, d\mathbf{x} &= \int_K \mathbf{r} \cdot \mathbf{q}_k \, d\mathbf{x} \quad \forall \mathbf{q}_k \in \mathbb{P}_k(K; \mathbb{R}^d) \setminus \mathbf{G}_k(K). \end{aligned}$$

Using the unsolvence of the degrees of freedom of $\mathbf{V}_K$, we deduce that $\mathbf{v}_h^{\text{div}} = \mathbf{r}$ on Ω . The boundary condition $\mathbf{v}_h^{\text{div}} \cdot \mathbf{n} = 0$ on $\partial\Omega$ then yields $\mathbf{r} \cdot \mathbf{n} = 0$ for all boundary normals $\mathbf{n}$. Since the boundary normals span $\mathbb{R}^d$, we get $\mathbf{r} = \mathbf{0}$. Therefore $\mathbf{v}_h^{\text{div}} = \mathbf{0}$ and, from $\mathbf{v}_h^t \cdot \boldsymbol{\tau} = \mathbf{r} \cdot \boldsymbol{\tau} = 0$ for all tangential $\boldsymbol{\tau}$, we also have $\mathbf{v}_h^t = \mathbf{0}$. Hence $\mathbf{v}_h = \mathbf{0}$, which proves that $\|\cdot\|_{1,h}$ is a norm on $\mathbf{V}_h$. $\square$

It follows from the Korn's inequality that

$$\|\mathbf{v}_h\|_{1,h,K}^2 \approx \|\nabla(\mathbf{\Pi}_K^o \mathbf{v}_h^{\text{div}})\|_K^2 + \|h^{-\frac{1}{2}} \mathbf{\Pi}_{\partial K} \mathcal{J}_{\partial K}(\mathbf{v}_h)\|_{\partial K}^2.$$

LEMMA 2.4. We have the discrete inf-sup condition

$$(2.8) \quad \|\mathbf{v}\|_{1,h} \lesssim \sup_{\boldsymbol{\sigma}_h \in \Sigma_h^{-1}} \frac{\sum_{K \in \mathcal{K}_h} (\epsilon_K^w(\mathbf{v}), \boldsymbol{\sigma})_K}{\|\boldsymbol{\sigma}\|_{0,h}} \quad \forall \mathbf{v} \in \mathbf{V}_h.$$

Proof. Let $\mathbf{v} \in \mathbf{V}_h$ be fixed. We construct a tensor $\boldsymbol{\sigma} \in \Sigma_h^{-1}$ satisfying the following moment conditions:

(i) For every $K \in \mathcal{K}_h$ and every $\boldsymbol{\tau} \in \mathbb{P}_{k-1}(K; \mathbb{S})$,

$$(\boldsymbol{\sigma}, \boldsymbol{\tau})_K = (\boldsymbol{\varepsilon}(\boldsymbol{\Pi}_K^o \mathbf{v}), \boldsymbol{\tau})_K.$$

(ii) For every face $F \in \mathcal{F}_h$ and every $\mathbf{q} \in V_k(F)$,

$$\langle \boldsymbol{\sigma} \mathbf{n}, \mathbf{q} \rangle_F = \langle h_F^{-1} \boldsymbol{\Pi}_{\partial K} \mathcal{J}_{\partial K}(\mathbf{v}), \mathbf{q} \rangle_F.$$

The DoFs (2.5b) vanish. By the unisolvence of the degrees of freedom for $\Sigma_h^{-1}(K)$ (Lemma 2.2), such a $\boldsymbol{\sigma}$ exists and is unique.

Using the definition of $\epsilon_K^w(\mathbf{v})$ and the above moment conditions, we compute

$$(\epsilon_K^w(\mathbf{v}), \boldsymbol{\sigma})_K = (\boldsymbol{\varepsilon}(\boldsymbol{\Pi}_K^o \mathbf{v}), \boldsymbol{\sigma})_K + (\boldsymbol{\Pi}_{\partial K} \mathcal{J}_{\partial K}(\mathbf{v}), \boldsymbol{\sigma} \mathbf{n})_{\partial K}.$$

By condition (i), the first term equals $\|\boldsymbol{\varepsilon}(\boldsymbol{\Pi}_K^o \mathbf{v})\|_K^2$. By condition (ii), the boundary term equals

$$\sum_{F \subset \partial K} h_F^{-1} \|\boldsymbol{\Pi}_{\partial K} \mathcal{J}_{\partial K}(\mathbf{v})\|_F^2.$$

Summing over all $K \in \mathcal{K}_h$, we obtain

$$\sum_{K \in \mathcal{K}_h} (\epsilon_K^w(\mathbf{v}), \boldsymbol{\sigma})_K = \sum_{K \in \mathcal{K}_h} \|\boldsymbol{\varepsilon}(\boldsymbol{\Pi}_K^o \mathbf{v})\|_K^2 + \sum_{K \in \mathcal{K}_h} \sum_{F \subset \partial K} h_F^{-1} \|\boldsymbol{\Pi}_{\partial K} \mathcal{J}_{\partial K}(\mathbf{v})\|_F^2 = \|\mathbf{v}\|_{1,h}^2.$$

On the other hand, a standard scaling argument together with the definition of the degrees of freedom yields

$$\|\boldsymbol{\sigma}\|_{0,h} \lesssim \|\mathbf{v}\|_{1,h}.$$

Therefore,

$$\begin{aligned} \|\mathbf{v}\|_{1,h}^2 &\lesssim \|\boldsymbol{\sigma}\|_{0,h} \sup_{\boldsymbol{\sigma}_h \in \Sigma_h^{-1}} \frac{\sum_{K \in \mathcal{K}_h} (\epsilon_K^w(\mathbf{v}), \boldsymbol{\sigma}_h)_K}{\|\boldsymbol{\sigma}_h\|_{0,h}} \\ &\lesssim \|\mathbf{v}\|_{1,h} \sup_{\boldsymbol{\sigma}_h \in \Sigma_h^{-1}} \frac{\sum_{K \in \mathcal{K}_h} (\epsilon_K^w(\mathbf{v}), \boldsymbol{\sigma}_h)_K}{\|\boldsymbol{\sigma}_h\|_{0,h}}. \end{aligned}$$

Dividing by $\|\mathbf{v}\|_{1,h}$ (if nonzero) gives the desired discrete inf-sup condition (2.8). The case $\mathbf{v} = \mathbf{0}$ is trivial. $\square$

As a corollary, we have the following lemma:

LEMMA 2.5. *For any $\mathbf{v}_h = \{\mathbf{v}_h^{\text{div}}, \mathbf{v}_h^t\} \in \mathbf{V}_h$, there holds that*

$$\|\mathbf{v}_h\|_{1,h}^2 \lesssim \|\epsilon_h^w(\mathbf{v}_h)\|^2.$$

Proof. By the discrete inf-sup condition established in Lemma 2.4, we have

$$\|\mathbf{v}_h\|_{1,h} \lesssim \sup_{\boldsymbol{\sigma} \in \Sigma_h^{-1}} \frac{\sum_{K \in \mathcal{K}_h} (\epsilon_K^w(\mathbf{v}_h), \boldsymbol{\sigma})_K}{\|\boldsymbol{\sigma}\|_{0,h}}.$$

For any $\boldsymbol{\sigma} \in \Sigma_h^{-1}$, the Cauchy–Schwarz inequality gives

$$\begin{aligned} \sum_{K \in \mathcal{K}_h} (\epsilon_K^w(\mathbf{v}_h), \boldsymbol{\sigma})_K &\leq \sum_{K \in \mathcal{K}_h} \|\epsilon_K^w(\mathbf{v}_h)\|_K \|\boldsymbol{\sigma}\|_K \\ &\leq \left(\sum_{K \in \mathcal{K}_h} \|\epsilon_K^w(\mathbf{v}_h)\|_K^2 \right)^{1/2} \left(\sum_{K \in \mathcal{K}_h} \|\boldsymbol{\sigma}\|_K^2 \right)^{1/2}. \end{aligned}$$

166 Since

$$167 \quad \|\epsilon_h^w(\mathbf{v}_h)\|^2 = \sum_{K \in \mathcal{K}_h} \|\epsilon_K^w(\mathbf{v}_h)\|_K^2, \quad \|\boldsymbol{\sigma}\|_{0,h}^2 = \sum_{K \in \mathcal{K}_h} \|\boldsymbol{\sigma}\|_K^2,$$

168 we obtain

$$169 \quad \sum_{K \in \mathcal{K}_h} (\epsilon_K^w(\mathbf{v}_h), \boldsymbol{\sigma})_K \leq \|\epsilon_h^w(\mathbf{v}_h)\| \|\boldsymbol{\sigma}\|_{0,h}.$$

170 Dividing by $\|\boldsymbol{\sigma}\|_{0,h}$ and taking the supremum over $\boldsymbol{\sigma} \in \Sigma_h^{-1}$, it follows that

$$171 \quad \sup_{\boldsymbol{\sigma} \in \Sigma_h^{-1}} \frac{\sum_{K \in \mathcal{K}_h} (\epsilon_K^w(\mathbf{v}_h), \boldsymbol{\sigma})_K}{\|\boldsymbol{\sigma}\|_{0,h}} \leq \|\epsilon_h^w(\mathbf{v}_h)\|.$$

172 Combining this with the discrete inf-sup condition yields

$$173 \quad \|\mathbf{v}_h\|_{1,h} \lesssim \|\epsilon_h^w(\mathbf{v}_h)\|.$$

174 Squaring both sides gives the desired discrete Korn inequality:

$$175 \quad \|\mathbf{v}_h\|_{1,h}^2 \lesssim \|\epsilon_h^w(\mathbf{v}_h)\|^2.$$

176 This completes the proof. $\square$

177 **3. Numerical scheme.** Let $S_h \subset L_0^2(\Omega)$ be a discontinuous piecewise P_{k-1}
 178 element space. Then, the corresponding discrete variational formulation for problem
 179 (1) is to seek $\mathbf{u}_h = \{\mathbf{u}_h^{\text{div}}, \mathbf{u}_h^t\} \in \mathbf{V}_h$ and $p_h \in S_h$ satisfying

$$180 \quad (14) \quad \begin{cases} a_h(\mathbf{u}_h, \mathbf{v}_h) + b(\mathbf{v}_h, p_h) = (\mathbf{f}, \boldsymbol{\Pi}_h^o \mathbf{v}_h^{\text{div}}) & \forall \mathbf{v}_h \in \mathbf{V}_h, \\ b(\mathbf{u}_h, q_h) = 0 & \forall q_h \in S_h, \end{cases}$$

181 where

$$182 \quad a_h(\mathbf{u}_h, \mathbf{v}_h) := (\epsilon_h^w(\mathbf{u}_h), \epsilon_h^w(\mathbf{v}_h)),$$

$$183 \quad b(\mathbf{v}_h, p_h) := -(\text{div } \mathbf{v}_h^{\text{div}}, p_h).$$

184 **LEMMA 3.1.** *For every $\mathbf{v}_h = \{\mathbf{v}_h^{\text{div}}, \mathbf{v}_h^t\} \in \mathbf{V}_h$, there holds*

$$185 \quad \|\epsilon_h^w(\mathbf{v}_h)\|^2 \lesssim \|\mathbf{v}_h\|_{1,h}^2.$$

186 *Proof.* Fix $K \in \mathcal{K}_h$. From the definition of the weak strain, for every $\boldsymbol{\sigma} \in \Sigma_h^{-1}(K)$,

$$187 \quad (\epsilon_K^w(\mathbf{v}_h), \boldsymbol{\sigma})_K = (\epsilon(\boldsymbol{\Pi}_K^o \mathbf{v}_h^{\text{div}}), \boldsymbol{\sigma})_K + \langle \boldsymbol{\Pi}_{\partial K} \mathcal{J}_{\partial K}(\mathbf{v}_h), \boldsymbol{\sigma} \mathbf{n} \rangle_{\partial K}.$$

188 Taking $\boldsymbol{\sigma} = \epsilon_K^w(\mathbf{v}_h)$, we obtain

$$189 \quad \|\epsilon_K^w(\mathbf{v}_h)\|_K^2 = (\epsilon(\boldsymbol{\Pi}_K^o \mathbf{v}_h^{\text{div}}), \epsilon_K^w(\mathbf{v}_h))_K + \langle \boldsymbol{\Pi}_{\partial K} \mathcal{J}_{\partial K}(\mathbf{v}_h), \epsilon_K^w(\mathbf{v}_h) \mathbf{n} \rangle_{\partial K}.$$

190 By the Cauchy–Schwarz inequality,

$$191 \quad \begin{aligned} & |(\epsilon(\boldsymbol{\Pi}_K^o \mathbf{v}_h^{\text{div}}), \epsilon_K^w(\mathbf{v}_h))_K| \\ & \leq \|\epsilon(\boldsymbol{\Pi}_K^o \mathbf{v}_h^{\text{div}})\|_K \|\epsilon_K^w(\mathbf{v}_h)\|_K \leq \|\epsilon(\boldsymbol{\Pi}_K^o \mathbf{v}_h^{\text{div}})\|_K \|\epsilon_K^w(\mathbf{v}_h)\|_K. \end{aligned}$$

192 For the boundary term, by the Cauchy–Schwarz inequality and the trace inequality
 193 for the finite-dimensional space $\Sigma_h^{-1}(K)$, there exists a constant $C_T > 0$, depending
 194 only on k and the shape regularity of $\mathcal{K}_h$, such that

$$195 \quad |\langle \Pi_{\partial K} \mathcal{J}_{\partial K}(\mathbf{v}_h), \epsilon_K^w(\mathbf{v}_h) \mathbf{n} \rangle_{\partial K}| \leq C_T \|h^{-1/2} \Pi_{\partial K} \mathcal{J}_{\partial K}(\mathbf{v}_h)\|_{\partial K} \|\epsilon_K^w(\mathbf{v}_h)\|_K.$$

196 Combining the above estimates gives

$$197 \quad \begin{aligned} \|\epsilon_K^w(\mathbf{v}_h)\|_K^2 &\leq \|\epsilon(\Pi_K^o \mathbf{v}_h^{\text{div}})\|_K \|\epsilon_K^w(\mathbf{v}_h)\|_K \\ &\quad + C_T \|h^{-1/2} \Pi_{\partial K} \mathcal{J}_{\partial K}(\mathbf{v}_h)\|_{\partial K} \|\epsilon_K^w(\mathbf{v}_h)\|_K. \end{aligned}$$

198 If $\|\epsilon_K^w(\mathbf{v}_h)\|_K \neq 0$, dividing by $\|\epsilon_K^w(\mathbf{v}_h)\|_K$ yields

$$199 \quad \|\epsilon_K^w(\mathbf{v}_h)\|_K^2 \lesssim \|\mathbf{v}_h\|_{1,h,K}^2.$$

200 The case $\|\epsilon_K^w(\mathbf{v}_h)\|_K = 0$ is trivial. This completes the proof. $\square$

201 **LEMMA 3.2.** *For all $\mathbf{u}_h, \mathbf{v}_h \in \mathbf{V}_h$, it holds that*

$$\begin{aligned} 202 \quad a_h(\mathbf{u}_h, \mathbf{v}_h) &\lesssim \|\mathbf{u}_h\|_{1,h} \|\mathbf{v}_h\|_{1,h}, \\ 203 \quad a_h(\mathbf{u}_h, \mathbf{u}_h) &\gtrsim \|\mathbf{u}_h\|_{1,h}^2. \end{aligned}$$

204 **LEMMA 3.3.** *For any $\mathbf{v}_h = \{\mathbf{v}_h^{\text{div}}, \mathbf{v}_h^t\} \in \mathbf{V}_h$ and $q_h \in S_h$, it is true that*

$$205 \quad b(\mathbf{v}_h, q_h) \lesssim \|\mathbf{v}_h\|_{1,h} \|q_h\|.$$

206 **LEMMA 3.4.** *For any $p_h \in S_h$, there exists a $\mathbf{v}_h = \{\mathbf{v}_h^{\text{div}}, \mathbf{v}_h^t\} \in \mathbf{V}_h$ such that*

$$207 \quad \text{div } \mathbf{v}_h^{\text{div}} = p_h \quad \text{and} \quad \|\mathbf{v}_h\|_{1,h} \lesssim \|p_h\|.$$

208 **3.1. The interpolation.** For any $\mathbf{v} \in \mathbf{H}^1(K)$, $\mathbf{I}_K^{\text{div}} \mathbf{v} \in \mathbf{V}_K$ is defined by the
 209 degree of freedoms, i.e.,

$$\begin{aligned} 210 \quad \int_F \mathbf{I}_K^{\text{div}} \mathbf{v} \cdot \mathbf{n} q_k &= \int_F \mathbf{v} \cdot \mathbf{n} q_k & \forall q_k \in P_k(F), F \subset \partial K, \\ 211 \quad \int_K \mathbf{I}_K^{\text{div}} \mathbf{v} \cdot \mathbf{q}_{k-2} &= \int_K \mathbf{v} \cdot \mathbf{q}_{k-2} & \forall \mathbf{q}_{k-2} \in \mathbf{G}_{k-2}(K), \\ 212 \quad \int_K \mathbf{I}_K^{\text{div}} \mathbf{v} \cdot \mathbf{q}_k &= \int_K \mathbf{v} \cdot \mathbf{q}_k & \forall \mathbf{q}_k \in \mathbf{P}_k(K) \setminus \mathbf{G}_k(K). \end{aligned}$$

213 **3.2. Error equations.** In this section, we assume that $(\mathbf{u}, p) \in (\mathbf{H}^{k+1} \cap \mathbf{H}_0^1) \times$
 214 $(H^k \cap L_0^2)$ is the weak solution of the Stokes problem, and define the interpolation
 215 $\mathbf{u}_I = \{\mathbf{u}_I^{\text{div}}, \mathbf{u}_I^t\}$ as $\mathbf{u}_I^{\text{div}}|_E = \mathbf{I}_E^{\text{div}} \mathbf{u}$, $\mathbf{u}_I^t|_{\partial E} = \Pi_{\partial E}^t \mathbf{u}$, $p_I = Q_h p$, where Q_h denotes the
 216 L^2 projection to the space S_h . We shall estimate $\mathbf{u}_h - \mathbf{u}_I$ and $p_h - p_I$.

217 Let $\mathcal{Q}_h : L^2(\Omega; \mathbb{S}) \rightarrow \Sigma_h^{-1}$ denote the L^2 -projection onto Σ_h^{-1} . For any $\mathbf{v}_h \in \mathbf{V}_h$,

218 we have

$$\begin{aligned}
& a_h(\mathbf{u}_h - \mathbf{u}_I, \mathbf{v}_h) + b(\mathbf{v}_h, p_h - p_I) \\
&= (\mathbf{f}, \Pi_K^o \mathbf{v}_h^{\text{div}}) - (\epsilon_h^w(\mathbf{u}_I), \epsilon_h^w(\mathbf{v}_h)) + (\nabla \cdot \mathbf{v}_h^{\text{div}}, Q_h p) \\
&= (-\nabla \cdot \epsilon(\mathbf{u}), \Pi_K^o \mathbf{v}_h^{\text{div}})_K + (\nabla p, \Pi_K^o \mathbf{v}_h^{\text{div}})_K \\
&\quad - (\epsilon(\Pi_K^o \mathbf{I}_K^{\text{div}} \mathbf{u}), \epsilon_h^w(\mathbf{v}_h))_K - \langle \Pi_{\partial K} \mathcal{J}_{\partial K}(\mathbf{u}_I), \epsilon_h^w(\mathbf{v}_h) \mathbf{n} \rangle_{\partial K} \\
&\quad + (\nabla \cdot \mathbf{v}_h^{\text{div}}, Q_h p) \\
&= (\epsilon(\mathbf{u}), \epsilon(\Pi_K^o \mathbf{v}_h^{\text{div}}))_K - \langle \epsilon(\mathbf{u}) \mathbf{n}, \Pi_K^o \mathbf{v}_h^{\text{div}} \rangle_{\partial K} \\
&\quad + (\nabla p, \Pi_K^o \mathbf{v}_h^{\text{div}})_K - (\mathbf{v}_h^{\text{div}}, \nabla Q_h p) + \langle Q_h p, \mathbf{v}_h^{\text{div}} \cdot \mathbf{n} \rangle_{\partial K} \\
&\quad - (Q_h \epsilon(\Pi_K^o \mathbf{I}_K^{\text{div}} \mathbf{u}), \epsilon(\Pi_K^o \mathbf{v}_h^{\text{div}})) - \langle \Pi_{\partial K} \mathcal{J}_{\partial K}(\mathbf{v}_h), Q_h \epsilon(\Pi_K^o \mathbf{I}_K^{\text{div}} \mathbf{u}) \mathbf{n} \rangle_{\partial K} \\
&\quad - \langle \Pi_{\partial K} \mathcal{J}_{\partial K}(\mathbf{u}_I), \epsilon_h^w(\mathbf{v}_h) \mathbf{n} \rangle_{\partial K} \\
219 \quad (3.1) \quad &= (\epsilon(\mathbf{u}) - Q_h \epsilon(\Pi_K^o \mathbf{I}_K^{\text{div}} \mathbf{u}), \epsilon(\Pi_K^o \mathbf{v}_h^{\text{div}}))_K \\
&\quad + (\Pi_K^o \mathbf{v}_h^{\text{div}}, \nabla p - \nabla Q_h p) \\
&\quad + \langle \epsilon(\mathbf{u}) \mathbf{n}, \mathcal{J}_{\partial K}(\mathbf{v}_h) \rangle_{\partial K} + \langle Q_h p - p, \mathbf{v}_h^{\text{div}} \cdot \mathbf{n} \rangle_{\partial K} \\
&\quad - \langle \Pi_{\partial K} \mathcal{J}_{\partial K}(\mathbf{v}_h), Q_h \epsilon(\Pi_K^o \mathbf{I}_K^{\text{div}} \mathbf{u}) \mathbf{n} \rangle_{\partial K} \\
&\quad - \langle \Pi_{\partial K} \mathcal{J}_{\partial K}(\mathbf{u}_I), \epsilon_h^w(\mathbf{v}_h) \mathbf{n} \rangle_{\partial K} \\
&= (\epsilon(\mathbf{u}) - Q_h \epsilon(\Pi_K^o \mathbf{I}_K^{\text{div}} \mathbf{u}), \epsilon(\Pi_K^o \mathbf{v}_h^{\text{div}}))_K \\
&\quad + \langle \mathcal{J}_{\partial K}(\mathbf{v}), \epsilon(\mathbf{u}) \mathbf{n} - Q_h \epsilon(\Pi_K^o \mathbf{I}_K^{\text{div}} \mathbf{u}) \mathbf{n} \rangle_{\partial K} \\
&\quad - (\nabla \cdot \Pi_K^o \mathbf{v}_h^{\text{div}}, p - Q_h p) + \langle Q_h p - p, \mathbf{v}_h^{\text{div}} \cdot \mathbf{n} - \Pi_K^o \mathbf{v}_h^{\text{div}} \cdot \mathbf{n} \rangle_{\partial K} \\
&\quad - \langle \Pi_{\partial K} \mathcal{J}_{\partial K}(\mathbf{u}_I), \epsilon_h^w(\mathbf{v}) \mathbf{n} \rangle_{\partial K}
\end{aligned}$$

220 Moreover, from the discrete divergence equation, for all $q \in S_h$,

$$221 \quad (3.2) \quad b(\mathbf{u}_h - \mathbf{u}_I, q) = 0.$$

222 We now estimate the terms on the right-hand side of (3.1). First,

$$\begin{aligned}
223 \quad & (\epsilon(\mathbf{u}) - Q_h \epsilon(\Pi_K^o \mathbf{I}_K^{\text{div}} \mathbf{u}), \epsilon(\Pi_K^o \mathbf{v}_h^{\text{div}})) \\
224 \quad & \leq \| \epsilon(\mathbf{u}) - Q_h \epsilon(\Pi_K^o \mathbf{I}_K^{\text{div}} \mathbf{u}) \| \| \epsilon(\Pi_K^o \mathbf{v}_h^{\text{div}}) \| \\
225 \quad & \leq \left(\| \epsilon(\mathbf{u}) - Q_h \epsilon(\mathbf{u}) \| + \| Q_h \epsilon(\mathbf{u}) - Q_h \epsilon(\Pi_K^o \mathbf{I}_K^{\text{div}} \mathbf{u}) \| \right) \| \mathbf{v}_h \|_{1,h} \\
226 \quad & \leq \left(Ch^k \| \mathbf{u} \|_{k+1} + \| \epsilon(\mathbf{u} - \Pi_K^o \mathbf{I}_K^{\text{div}} \mathbf{u}) \| \right) \| \mathbf{v}_h \|_{1,h} \\
227 \quad & \leq \left(Ch^k \| \mathbf{u} \|_{k+1} + \| \epsilon(\Pi_K^o (\mathbf{u} - \mathbf{I}_K^{\text{div}} \mathbf{u})) \| \right) \| \mathbf{v}_h \|_{1,h} \\
228 \quad & \leq Ch^k \| \mathbf{u} \|_{k+1} \| \mathbf{v}_h \|_{1,h}.
\end{aligned}$$

229 Second,

$$\begin{aligned}
230 \quad & \langle \mathcal{J}_{\partial K}(\mathbf{v}_h), \epsilon(\mathbf{u}) \mathbf{n} - Q_h \epsilon(\Pi_K^o \mathbf{I}_K^{\text{div}} \mathbf{u}) \mathbf{n} \rangle_{\partial K} \\
231 \quad & \leq h^{-1/2} \| \mathcal{J}_{\partial K}(\mathbf{v}_h) \|_{\partial K} \cdot h^{1/2} \| \epsilon(\mathbf{u}) - Q_h \epsilon(\Pi_K^o \mathbf{I}_K^{\text{div}} \mathbf{u}) \|_{\partial K} \\
232 \quad & \leq Ch^k \| \mathbf{u} \|_{k+1} \| \mathbf{v}_h \|_{1,h}.
\end{aligned}$$

Third,

$$\begin{aligned} & (\nabla \cdot \Pi_K^o \mathbf{v}_h^{\text{div}}, p - Q_h p) \\ & \leq \|\nabla \cdot \Pi_K^o \mathbf{v}_h^{\text{div}}\| \|p - Q_h p\| \\ & \leq Ch^k \|p\|_k \|\mathbf{v}_h\|_{1,h}. \end{aligned}$$

Fourth,

$$\begin{aligned} & \langle Q_h p - p, \mathbf{v}_h^{\text{div}} \cdot \mathbf{n} - \Pi_K^o \mathbf{v}_h^{\text{div}} \cdot \mathbf{n} \rangle_{\partial K} \\ & \leq \|Q_h p - p\|_{\partial K} \|(\mathbf{v}_h^{\text{div}} - \Pi_K^o \mathbf{v}_h^{\text{div}}) \cdot \mathbf{n}\|_{\partial K} \\ & \leq Ch^k \|p\|_k \|\mathbf{v}_h\|_{1,h}. \end{aligned}$$

Finally,

$$\begin{aligned} & \langle \Pi_{\partial K} \mathcal{J}_{\partial K}(\mathbf{u}_I), \epsilon_h^w(\mathbf{v}_h) \mathbf{n} \rangle_{\partial K} \\ & \leq \|\Pi_{\partial K} \mathcal{J}_{\partial K}(\mathbf{u}_I)\|_{\partial K} \|\epsilon_h^w(\mathbf{v}_h)\|_{\partial K} \\ & \leq h^{-1/2} \|\Pi_{\partial K} \mathcal{J}_{\partial K}(\mathbf{u}_I)\|_{\partial K} \|\mathbf{v}_h\|_{1,h}, \end{aligned}$$

and

$$\begin{aligned} & h^{-1} \|\Pi_{\partial K} \mathcal{J}_{\partial K}(\mathbf{u}_I)\|_{\partial K}^2 \\ & \leq h^{-1} \left(\|\mathbf{I}_K^{\text{div}} \mathbf{u} - \Pi_K^o \mathbf{I}_K^{\text{div}} \mathbf{u}\|_{\partial K}^2 + \|\mathbf{u} - \Pi_K^o \mathbf{I}_K^{\text{div}} \mathbf{u}\|_{\partial K}^2 \right) \\ & \leq (Ch^k \|\mathbf{u}\|_{k+1})^2. \end{aligned}$$

Hence,

$$\langle \Pi_{\partial K} \mathcal{J}_{\partial K}(\mathbf{u}_I), \epsilon_h^w(\mathbf{v}_h) \mathbf{n} \rangle_{\partial K} \leq Ch^k \|\mathbf{u}\|_{k+1} \|\mathbf{v}_h\|_{1,h}.$$

Combining the above estimates yields the desired error bound.

3.3. Error estimates. Let $e_h = \mathbf{u}_h - \mathbf{u}_I$ and $\xi_h = p_h - p_I$. Taking $\mathbf{v}_h = e_h$ in the error equation and using the discrete divergence constraint $b(e_h, q_h) = 0$ for all $q_h \in S_h$ with $q_h = \xi_h$, we obtain

$$a_h(e_h, e_h) = \mathcal{R}(e_h),$$

where $\mathcal{R}(\mathbf{v}_h)$ denotes the right-hand side of the error equation. From the coercivity of a_h , we have

$$\|e_h\|_{1,h}^2 \lesssim a_h(e_h, e_h) = \mathcal{R}(e_h).$$

Using the estimates for the terms of $\mathcal{R}(\mathbf{v}_h)$ derived above, we get

$$|\mathcal{R}(\mathbf{v}_h)| \lesssim h^k (\|\mathbf{u}\|_{k+1} + \|p\|_k) \|\mathbf{v}_h\|_{1,h}.$$

Taking $\mathbf{v}_h = e_h$, it follows that

$$\|e_h\|_{1,h} \lesssim h^k (\|\mathbf{u}\|_{k+1} + \|p\|_k).$$

Next, we estimate the pressure error. By the discrete inf-sup condition, for any $q_h \in S_h$, there exists $\mathbf{v}_h \in \mathbf{V}_h$ such that

$$b(\mathbf{v}_h, q_h) \gtrsim \|\mathbf{v}_h\|_{1,h} \|q_h\|.$$

266 From the error equation, we have

$$267 \quad b(\mathbf{v}_h, \xi_h) = -a_h(e_h, \mathbf{v}_h) + \mathcal{R}(\mathbf{v}_h).$$

268 Thus,

$$269 \quad \|\xi_h\| \lesssim \sup_{\mathbf{v}_h \in \mathbf{V}_h} \frac{-a_h(e_h, \mathbf{v}_h) + \mathcal{R}(\mathbf{v}_h)}{\|\mathbf{v}_h\|_{1,h}} \lesssim \|e_h\|_{1,h} + h^k(\|\mathbf{u}\|_{k+1} + \|p\|_k).$$

270 Combining this with the estimate for $\|e_h\|_{1,h}$, we obtain

$$271 \quad \|\xi_h\| \lesssim h^k(\|\mathbf{u}\|_{k+1} + \|p\|_k).$$

272 Finally, by the triangle inequality and the interpolation error estimates,

$$273 \quad \|\mathbf{u} - \mathbf{u}_h\|_{1,h} \leq \|\mathbf{u} - \mathbf{u}_I\|_{1,h} + \|e_h\|_{1,h} \lesssim h^k(\|\mathbf{u}\|_{k+1} + \|p\|_k),$$

274 and

$$275 \quad \|p - p_h\| \leq \|p - p_I\| + \|\xi_h\| \lesssim h^k(\|\mathbf{u}\|_{k+1} + \|p\|_k).$$

276 Therefore, we have the optimal error estimate

$$277 \quad \|\mathbf{u} - \mathbf{u}_h\|_{1,h} + \|p - p_h\| \lesssim h^k(\|\mathbf{u}\|_{k+1} + \|p\|_k).$$